

UC Berkeley researchers open source Reinforcement Learning with Augmented Data

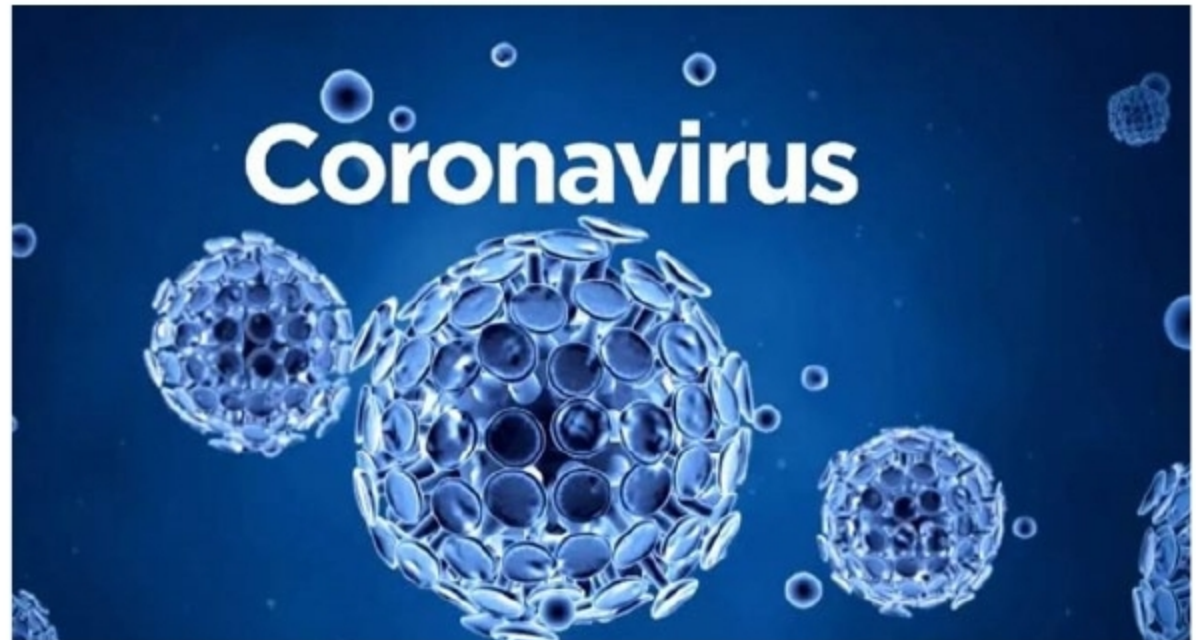
As per a report by *Venturebeat*, a group of researchers from the University of California, Berkeley, have open sourced Reinforcement Learning with Augmented Data (RAD). In an accompanying paper, the authors said that this module can improve any existing reinforcement learning algorithm, and RAD can achieve better compute and data efficiency than Google AI's PlaNet and other cutting-edge algorithms like DeepMind's Dreamer and SLAC from UC Berkeley.

The researchers said that RAD achieved state-of-art results on common benchmarks and matched or beat every baseline in terms of performance and data efficiency across 15 DeepMind control environments. It added that it did this in part by applying data augmentations for visual observations. The co-authors of the paper on RAD include Michael 'Misha' Laskin, Kimin Lee, and Berkeley AI Research co-director and Covariant founder Pieter Abbeel. As per the report, RAD was released on preprint repository arXiv.

The paper said that, for the first time, it can be shown that data augmentations alone can significantly improve the data-efficiency and generalisation of RL methods operating from pixels. This can be done without any changes to the underlying RL algorithm, on the DeepMind Control Suite and the OpenAI ProcGen benchmarks, respectively. It added that by using multiple augmented views of the same data point as input, CNNs are forced to learn consistencies in their internal representations. This results in a visual representation that improves generalisation, data-efficiency, and transfer learning.

who have doctoral degrees in machine learning and immunobiology, bioinformaticians, and Web developers.

The process of developing a vaccine is typically a very lengthy and costly one due to the number of virus-infected cells that need to be analysed. Dr Daouda has developed an AI algorithm that can predict which parts of a virus are more likely to be exposed at the surface of infected cells, which are called epitopes. He did this while obtaining his doctorate degree at the Institute for Research in Immunology and Cancer at the Université de Montréal.



These predictions can be used by researchers to generate a significantly shorter list of potential targets to test in the creation of a vaccine. It reduces a process that typically takes weeks or months, to just hours.

Authentic8 launches Open Source Intelligence Academy

Authentic8 has launched the Open Source Intelligence Academy (OSINT), which will have an integrated suite of resources and tools available exclusively to its customers. It aims to augment their training securely from home and other remote

work environments. The resources will be delivered within Silo, Authentic8's Web isolation platform.

OSINT Academy will include the OSINT Insider talk webcast series, the OSINT Forum for collaboration and peer-to-peer exchange, analysis case studies, demonstration videos, and OSINT Tradecraft Training. The academy will provide access to a range of resources designed to get new analysts up to speed, streamline their intelligence cycle, and equip



experienced researchers with new skills to stay current.

The curriculum was developed in partnership with training specialist OSINT Combine to create more than 40 targeted training modules. It includes videos, presentations, study guides and interactive quiz materials designed to build and test analysts' skillsets.

SUSE joins hands with iValue to offer open source solutions for enterprises

Technology aggregator iValue InfoSolutions has announced that it is partnering with SUSE to offer enterprise-grade, open source solutions for Linux, software-